

Gaussian filter

Write a function **GaussianFilter** that takes as input a [1xN] vector **x**, which represents a signal, and a scalar **sigma**. Use the 'conv' function with standard deviation **sigma**. Choose **W**, the size of the filter, to be $6 \times \text{sigma}$, rounding up using 'ceil' to ensure the result is an integer. The filter **y**. Use the 'same' argument with 'conv' to ensure the output **y** is the same size as the input **x**.

The steps are:

- Compute W by rounding up $6 \times \text{sigma}$
- Use fspecial to create our Gaussian filter
- Create output signal by convolving the signal with the filter

The Gaussian filter smears out non-zero amplitudes in **x**. As an example, when **x** contains only 1 non-zero entry, that single value gets smeared out over the entries in the array:

```
x = [0 0 0 1 0 0 0];
sigma = 1.5;
y = GaussianFilter(x, sigma)
```

Gives the result

```
y = [0.0361    0.1096    0.2134    0.2666    0.2134    0.1096    0.0361]
```

Function ?

 Save

 Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function y = GaussianFilter(x, sigma)
2 % Compute W by rounding up 6*sigma
3 W = round(6*sigma);
4 % Use fspecial to create our Gaussian filter
5 filt = fspecial("gaussian", [W, 1], sigma);
6 % convolve the signal with the filter
7 y = conv(x, filt, 'same');
8 end
```

Code to call your function ?

 Reset

```
1 x = [0 0 0 1 0 0 0];
2 sigma = 1.5;
3 y = GaussianFilter(x, sigma)
```

 Run Function



Assessment: All Tests Passed

Submit ?

 Is output y correct for random valued signal?